

Gender-Based Price Differences: The Case of Deodorants

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Disclaimer

Researcher(s)' own analyses calculated (or derived) based in part on data from Nielsen Consumer LLC and marketing databases provided through the NielsenIQ Datasets at the Kilts Center for Marketing Data Center at The University of Chicago Booth School of Business.

The conclusions drawn from the NielsenIQ data are those of the researcher(s) and do not reflect the views of NielsenIQ. NielsenIQ is not responsible for, had no role in, and was not involved in analyzing and preparing the results reported herein.

Introduction

- In recent years, the term “pink tax” gained notoriety.
 - ▶ Higher prices for female-oriented than male-oriented goods.
 - ▶ Under similar characteristics (ingredients, packaging, producer, etc.).
- There's concern about female consumers being harmed by firms.
 - ▶ Some governments implemented legislation to fight it.
 - ▶ Miami (1993), California (1995 and 2024) and New York (2020).
- No clarity on what can cause this gender-based differences.
 - ▶ Literature has focused on own-price elasticities and production costs.
 - ▶ None of these seem to explain the pink-tax.
 - ▶ However, little attention has been posed into substitution effects.

Introduction



Research Questions

1. Is there actually a pink tax?
 2. In case there is, what can explain this?
 3. What are the welfare effects of gender-parity policies?
- Using linear regression I try to answer question (1)
 - ▶ Replicate a strategy by Moshary et al. (2023) with some aggregates.
 - For (2), literature has missed 2 relevant elements explaining the gap.
 - ▶ (i) Costs and (ii) portfolio.
 - ▶ Structural model to decompose the pink-tax.
 - Use the structural model to answer (3).
 - ▶ Solve equilibrium prices under gender-based constraints.
 - Focus on the market of deodorants.

Summary of Results

- Reduced-form:
 - ▶ Controlling for observables, female-oriented prices tend to be higher.
- Using the structural model I decompose the price gap:
 - ▶ A small part is explained by own-price sensitivity.
 - ▶ Most of the pink-tax is explained by costs.
 - ▶ Once costs are controlled for, the pink-tax almost disappears.
- How does a gender-parity policy affect welfare?
 - ▶ Considering all products: no diff. between male and female consumers.
 - ▶ Considering gendered goods: male are better- and female are worse-off.
 - ▶ Big firms have higher profits, small firms have lower profits.

- **Gender-based price discrimination:** Moshary, Tuchman & Vajravelu (2023); Guittar et al. (2022); Duesterhaus (2011); Ayres (1991); Ayres & Siegelman (1995); Bessendorf (2015).
- **Differentiated products' choice:** Berry, Levinsohn and Pakes, (1995, 2004), Nevo (2001), Conlon and Gortmaker (2020), DellaVigna and Gentzkow (2019), Gandhi and Houde (2019).
- **Gender + differentiated products:** Barnes and Brounstein (2023).
- Main contribution of this paper: None of these papers
 - ▶ Provides evidence on the source of pink-tax.
 - ▶ Analyzes potential effects of gender-parity policies.

- **NielsenIQ RSD:**

- ▶ 90 retail chains (35-50k stores) in the US
- ▶ 2.5-4.5 million of products in weekly sales.

- **NielsenIQ CSD:**

- ▶ 40,000-60,000 households across the US
- ▶ Focus on SP households, use to categorize goods in terms of gender.

- **Competitive Promotion Report:**

- ▶ Data on producers' wholesale prices at national level.

- **Label Insight:**

- ▶ Data on products' packaging text, used to extract ingredients.

Data and Definitions

- Markets are defined at the Retailer-DMA-Month.
 - ▶ Market size estimated using dairy products as in Conlon et al. (2023).
 - ▶ Year 2018 in the state of New York.
- Products are defined at the band-gender-ingredients level per Oz.
 - ▶ Dove products targeting men and having the same first 5 ingredients.
 - ▶ Price is the price per Oz, while units is the number of Oz.
- Gender of products is defined using:
 1. Use GPT-AI to classify brands (e.g., Old Spice). GPT Example
 2. If (1) is not possible, use CPD (Barnes and Brounstein, 2023).

Descriptive Statistics

	Female	Male	Non-gender
# of products	103	167	100
# of firms	8	8	9
% of total sales	35.8%	40.3%	23.8%
% of products with ethanol	48.5%	52.0%	44.0%
% of products with aluminum	17.5%	12.0%	25.0%
% of products with organic ingredients	1.0%	0.0%	5.0%

- More male-oriented products.
- Salient differences in ingredients:
 - ▶ More non-gender goods with aluminum and organic ingredients

Descriptive Statistics

	Female		Male		Non-gendered	
	mean	s.d.	mean	s.d.	mean	s.d.
price	5.05	2.05	4.96	1.78	4.61	2.24
price/Oz	2.11	1.25	1.75	1.01	1.75	1.01
Size	2.91	2.37	4.21	11.55	2.85	0.96
Sales (monthly)	161.21	739.23	127.87	473.52	135.36	747.01
Wholesale price	3.67	1.39	3.48	1.46	3.15	1.34
Wholesale price/Oz	1.51	0.79	1.22	0.68	1.18	0.59

- Female products more expensive per unit and with more sales.

	All	Female	Male	Non-gender
Unilever	33.8	44.6	39.6	59.0
P&G	45.8	43.7	41.8	1.7
Colgate Palmolive	6.3	3.2	8.8	7.5
Church and Dwight	3.4	0.1	0.2	14.8
Henkel	3.0	0.0	6.2	2.5

- Highly concentrated market.

Reduced-form Evidence on Pink Tax

- Replicate strategy by Moshary et al. (2023) with 2 aggregates:
 - ▶ Allow for non-gendered goods. Example
 - ▶ Check for wholesale prices.
- Run the following regression:

$$p_{imt} = \beta_{male} male_{it} + \beta_{non} non_{it} + X_i' \alpha + \gamma_{mt} + \gamma_f + \epsilon_{imt} \quad (1)$$

- With:
 - ▶ $male_{it}$ and non_{it} : dummy vars for male and non-gender goods.
 - ▶ X_i' ingredient variables: first 5 ingred., aluminum, alcohol and organic.
 - ▶ γ_{mt} and γ_f : market and firm fixed effects.

Reduced-form Evidence on Pink Tax: Retail Prices

	(1)	(2)	(3)
β_{male}	-0.363*** (0.007)	-0.279*** (0.007)	-0.064*** (0.005)
β_{non}	-0.371*** (0.008)	-0.086*** (0.009)	-0.013*** (0.005)
N	117,857	117,857	117,857
R^2	0.049	0.126	0.782
Market	Y	Y	Y
Manuf.	N	Y	Y
Ing.	N	N	Y

clustered s.e. in parentheses, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

- Not controlling for ingredients brings a large gap.
 - ▶ Once ingredients are accounted for, gap reduced in around \$0.06.
- Average size of 3 Oz $\rightarrow$ average gap of $\sim$ \$0.20.

Reduced-form Evidence on Pink Tax: Wholesale Prices

	(1)	(2)	(3)
β_{male}	-0.326*** (0.031)	-0.210*** (0.031)	-0.088*** (0.021)
β_{non}	-0.228*** (0.035)	-0.030 (0.044)	-0.023 (0.036)
N	3,927	3,927	3,920
R^2	0.246	0.337	0.818
Market	Y	Y	Y
Manuf.	N	Y	Y
Ing.	N	N	Y

clustered s.e. in parentheses, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

- Gap is also present at wholesale level.
 - ▶ Once ingredients are accounted for, gap reduced in around \$0.09.
- Average size of 3 Oz $\rightarrow$ average gap of $\sim$ \$0.27.

Reduced-form Evidence on Pink Tax: Results

- Previous results indicate 2 conclusions.
 1. Female deodorants are more expensive than male deodorants.
 2. Gap is even higher with non-gendered goods.
- Problem may look more dramatic if quality is not controlled for.
- To better measure of the gender gap we need to account for costs:
 - ▶ Recover marginal costs from demand estimation and use it as control.

Structural Model: Demand Side

- Use a RC framework (Berry, Levinsohn & Pakes, 1995), with utility:

$$\begin{aligned} U_{ijt} = & \alpha p_{jt} + \beta_f fem_j + \beta_m male_j + \beta_e alc_j + \beta_a alum_j + \beta_s size_j \\ & \beta_{f \times e} fem_j \times alc_j + \beta_{m \times e} mal_j \times alc_j + \beta_{f \times a} fem_j \times alum_j \\ & + \beta_{m \times a} mal_j \times alum_j + \sigma_0 v_i + \sigma_{price} v_i p_{jt} + \gamma_{B(j)} \times \gamma_{I(j)} + \\ & \gamma_t + \xi_{jt} + \epsilon_{ijt} \end{aligned} \quad (2)$$

- Heterogeneous tastes (μ_{ijt}) are modelled as:

$$\mu_{ijt} = \sigma_0 v_i + \sigma_{price} v_i p_{jt} \quad (3)$$

- With T1EV errors, markets shares can be expressed as:

$$s_{ijt}(\delta_{ijt}) = \int \frac{\exp(\delta_{jt} + \mu_{ijt})}{\sum_k \exp(\delta_{kt} + \mu_{ikt})} d\mu_{it} \quad (4)$$

Structural Model: Demand Side

- Estimate by GMM using `pyblp` by Conlon and Gortmaker (2020).
- Use instruments proposed by Gandhi and Houde (2019).
 - ▶ Difference in characteristics of goods in a market.
 - ▶ Given a characteristic l for of good j : $d_{jt}^l = |z_{jt}^k - z_{j't}^k|$.
 - ▶ Use the mean difference of j with all goods $j' \in \mathcal{J}_m, j \neq j'$.
 - ▶ k : aluminum, alcohol and size.
- X : female and male dummy, brand fixed effect, ingredient fixed effect.

Structural Model: Supply Side

- Set of multi-prod firms $\mathcal{F}$, each $f \in \mathcal{F}$ sells $\mathcal{J}_f^t$ goods in t .
- Firms compete in prices, s.t. for any $j \in \mathcal{J}_f^t$, firm f solves:

$$\max_{p_j} \pi(p_{1t}, \dots, p_{Jt}) = \sum_{k \in \mathcal{J}_f^t} p_{kt} Q_{kt}(p_{1t}, \dots, p_{Jt}) - C(Q_{kt}(p_{1t}, \dots, p_{Jt})) \quad (5)$$

- FOC for all firms in t can be expressed as:

$$\mathbf{Q}_t + \Omega \Delta_d [\mathbf{p}_t - \mathbf{mc}_t] = 0 \quad (6)$$

- Where:

- ▶ Ω is a $\mathcal{J}_j^t \times \mathcal{J}_j^t$ ownership matrix.
- ▶ Δ_d is a $\mathcal{J}_j^t \times \mathcal{J}_j^t$ price-sensitivity matrix, s.t., $\Delta_{d(k,l)} = \frac{dQ_{kt}}{dp_{lt}}$.
- ▶ $\mathbf{Q}_t$ and $\mathbf{p}_t$ are $\mathcal{J}_j^t \times 1$ vectors of data, $\mathbf{mc}_t$ can be recovered.

Demand Parameter Estimates

α	-2.4586*** (0.4335)
β_f	-0.4469*** (0.0741)
β_m	-1.0566*** (0.1067)
β_a	0.1919** (0.0890)
β_e	-0.2637** (0.1069)

β_s	-1.2552** (0.3421)
$\beta_{f \times a}$	-0.6404*** (0.0420)
$\beta_{m \times a}$	0.7389*** (0.1158)
$\beta_{f \times e}$	0.6502*** (0.0557)
$\beta_{m \times e}$	0.3132*** (0.0731)
σ_0	1.011*** (0.0001)
σ_{price}	0.0001 (1.5420)
Median elasticity	-3.885
Number of Firms	10
Number of Markets	848
N	117,624

Clustered s.e. at the market level in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Counterfactuals

- Want to understand why we observe a gender gap.
- Three elements affect pricing decisions:
 1. Own-price sensitivity of demand.
 2. Marginal costs.
 3. Substitution/cannibalization.

$$p_j = \underbrace{-\left(\frac{\partial Q_j}{\partial p_j}\right)^{-1} Q_j}_1 + \underbrace{c_j}_2 + \underbrace{\left(\frac{\partial Q_j}{\partial p_j}\right)^{-1} \left[\sum_{k \in \mathbf{J}_f, k \neq j} (p_k - c_k) \left(\frac{\partial Q_k}{\partial p_j}\right) \right]}_3$$

- If c for female goods are higher, ceteris paribus, prices will be higher.
- Propose counterfactual to shut down these effects.
 - ▶ Check price gaps under these scenarios.

Case 1: No Substitution

- Female goods can have more or stronger substitutes than male ones.
 - ▶ When $\uparrow p_j$, female consumers are very likely to buy $j' \in \mathcal{J}_f$, or
 - ▶ When $\uparrow p_j$, female consumers have more substitutes in $\mathcal{J}_f$ than male.
- Suppose the substitution effect is the same across products: 0.
- Equilibrium prices are:

$$p_j = - \left(\frac{\partial Q_j}{\partial p_j} \right)^{-1} Q_j + c_j \quad (7)$$

- If there is a pink-tax it must come from:
 - ▶ Higher cost for female products.
 - ▶ More sensitive demand.

Case 2: No Difference in Costs

- If marginal costs are higher for female goods, prices will be higher.
- Suppose there is no difference in costs:
 - ▶ Firm f sells good j and k , s.t. $female_j = 1$ and $male_k = 1$.
 - ▶ And $ingredients_j = ingredients_k$.
 - ▶ Then $\bar{c}_j = \bar{c}_k$, and make this cost $\bar{c}_j = \bar{c}_k = \frac{c_j + c_k}{2}$
 - ▶ Where $\bar{c}$ is the counterfactual cost.
- Here, if there is a pink tax it must come from:
 - ▶ More relevant substitution/cannibalization.
 - ▶ More sensitive demand.

Case 3: No Substitution + No Difference in Costs

- Mix of Case 1 and Case 2.
- Prices are:

$$p_j = - \left(\frac{\partial Q_j}{\partial p_j} \right)^{-1} Q_j + \bar{c}_j \quad (8)$$

- The only explanation for a pink-tax:
 - ▶ A more sensitive demand for female goods.

Pink Tax Under Counterfactual Scenarios

	(1) Baseline	(2) No substitution	(3) No cost gap	(4) Both
β_{male}	-0.0636*** (0.0046)	-0.0611*** (0.0045)	-0.0123* (0.0066)	-0.0140*** (0.0046)
β_{non}	-0.0140** (0.0055)	-0.0135** (0.0054)	0.0336*** (0.0079)	0.0324*** (0.0055)
Market	Y	Y	Y	Y
Manuf.	Y	Y	Y	Y
Ing.	Y	Y	Y	Y
N	117920	117920	117920	117920
R^2	0.776	0.769	0.723	0.764

- Shutting down substitution does not affect the pink-tax.
- If no difference in costs, there will still be a pink-tax \$0.01.

Pink Tax Under Counterfactual Scenarios

	(1) Baseline	(2) No substitution	(3) No cost gap	(4) Both
β_{male}	-0.0001 (0.0002)	-0.0002*** (0.0001)	0.0312*** (0.0023)	0.0349*** (0.0024)
β_{non}	-0.0012*** (0.0002)	-0.0002** (0.0001)	0.0396*** (0.0028)	0.0442*** (0.0029)
mc	1.005*** (0.0001)	1.006*** (0.00003)	0.696*** (0.0015)	0.676*** (0.0016)
Market	Y	Y	Y	Y
Manuf.	Y	Y	Y	Y
Ing.	Y	Y	Y	Y
N	117920	117920	117920	117920
R^2	0.910	0.889	0.867	0.845

- Once marginal costs are controlled for, there is no pink tax.
- If costs were the same, male goods would be more expensive.

Gender-parity Policy

- In 2020 NY banned the possibility of discriminate based on gender:

“(...) any individual or entity, including retailers, suppliers, manufacturers or distributors, are prohibited from charging a price for two ”substantially similar” goods or services, if the goods or services are priced differently based on the gender for whom the goods or services are marketed. (...)”
(Department of State, 2020)

- Hard to evaluate this policy:

- ▶ Prices change in NY, nearby states and wholesalers. Wholesale Prices
- ▶ There is a relative increase in male (relative) prices.
- ▶ Historical peak of input prices in that time. Input Prices

- No enforcement? hard to enforce?

How Good is Gender-parity Policy?

- Not trivial that women will be strictly better-off:
 - ▶ In some markets, male-oriented goods may be more expensive.
 - ▶ Women also buy non-gender products, and those prices can go up.
- Use the previous structural model to solve the problem:

$$\max_{p_j} \pi(p_{1t}, \dots, p_{Jt}) = \sum_{k \in \mathcal{J}_f^t} p_{kt} Q_{kt}(p_{1t}, \dots, p_{Jt}) - C(Q_{kt}(p_{1t}, \dots, p_{Jt})) \quad (9)$$

$$\text{s.t. } p_{ht} = p_{gt} \text{ if } \begin{cases} X_h = X_g \text{ and } female_h = male_g = 1, |c_h - c_g| < 0.5, \text{ or} \\ X_h = X_g \text{ and } male_h = female_g = 1, |c_h - c_g| < 0.5 \end{cases}$$

- Compute welfare changes for two cases.
 1. Consumers buy goods from their gender + non-gendered goods.
 2. Consumers buy only gendered products.

Welfare Changes: Female Consumers

	(1) $\Delta\text{Welfare}$	(2) $\% \text{ mkts w/ } \Delta\text{Welfare} < 0$	(3) $\Delta\text{Welfare (gendered)}$
Men	1.8%	35.7%	1.0%
Women	1.6%	37.4%	0.0%
Big Firms	52.9%	13.3%	-
Small Firms	13.9%	59.0%	-

- With non-gendered purchases, both male and female are better-off.
- With only gendered-goods, male are better- and female are worse-off.
 - ▶ We are conditioning on same marginal costs.
 - ▶ As seen before, conditional on costs male products are more expensive.
 - ▶ Female goods' prices increase in 50% markets, male in 36%.
- Big firms find higher profits in most markets, but not small firms.
 - ▶ May allow for an equilibrium not possible under pure price competition.

Conclusions

- In deodorants: a gap in prices based on gender of around \$0.07/Oz.
 - ▶ Most of this gap comes from differences in costs.
 - ▶ Gap is not caused by differences in price-sensitivity.
 - ▶ Artificially eliminating the gap: women improve welfare more than men.
- What happens if firms perfectly follow the pink-tax ban?
 - ▶ Women find higher welfare in 90% of markets, lower welfare in 10%.
 - ▶ Some undesired effects!
- For firms:
 - ▶ Big firms increase profits, the rest win and lose in some markets.
 - ▶ Intuition: legislation can allow for a better equilibrium.
 - ▶ This could rationalize part of the price change post-legislation.
- Own-demand price sensitivity is not important.
 - ▶ Price gap not coming from women having a more inelastic demand.

GPT Example

I am doing research on the demand for deodorants in the U.S. and need help identifying products' names and their targeted gender. I will proceed to give you a list of brands and need your help with one task. Using the brand name I need you to tell me the gender it targets (i.e., male, female or non-gendered). If you are not sure about the gender, please say it's non-gendered. Please, display the output in the following way: ['brand name'; 'gender'].

- I later repeat this for product descriptions for brands that could not be classified.
- The data fed into the AI tool comes from product descriptions in CPR data, not NielsenIQ.

Gendered Brands

Female brands	Male Brands
CRYSTAL	AOS ART OF SPORT
DRY IDEA	AXE
ELIZABETH ARDEN	BEVEL
LOVE BEAUTY AND PLANET	BRUT
MEGABABE	DR. SQUATCH
SECRET	DOVE MEN
SEVENTH GENERATION	DUKE CANNON SUPPLY CO.
SOFT & DRI	EVERY MAN JACK
LADY SPEED STICK	GILLETTE
	MENNEN
	OBSESSION FOR MEN
	OLD SPICE
	RIGHT GUARD
	SPEED STICK
	CK
	CURVE

Example: P&G Deodorants



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Aluminum Prices



source: tradingeconomics.com

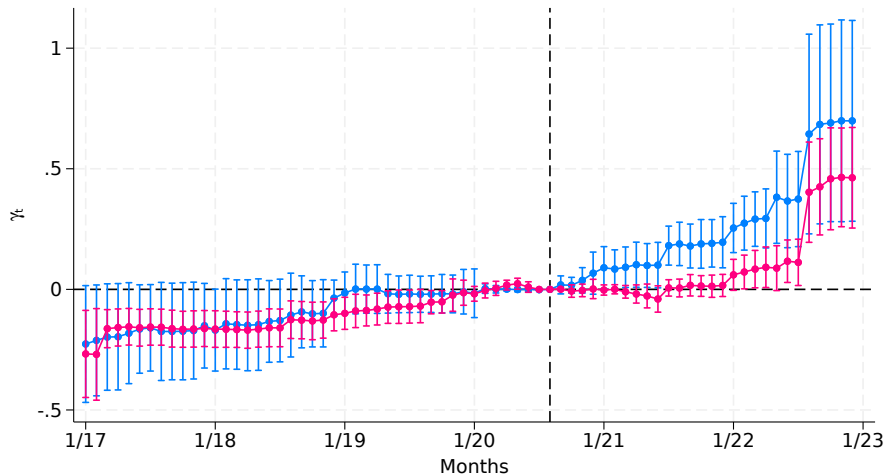
Ethanol Prices



source: tradingeconomics.com

Wholesale Price Trend

Wholesale Price Evolution



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Pink-tax Measure

	CT (1)	MD (2)	MA (3)	NJ (4)	NY (5)	PA (6)	VT (7)
(a) Pre-legislation							
β_{male}	-0.039*** (-8.39)	-0.049*** (-15.75)	-0.042*** (-13.04)	-0.080*** (-19.48)	-0.048*** (-21.65)	-0.051*** (-20.17)	-0.027*** (-5.53)
β_{non}	-0.067*** (-12.06)	-0.050*** (-13.31)	-0.065*** (-17.10)	-0.072*** (-15.25)	-0.085*** (-33.17)	-0.093*** (-30.90)	-0.078*** (-13.43)
R^2	0.839	0.832	0.829	0.831	0.825	0.826	0.836
N	103293	219414	209065	142231	491875	344213	82900
(b) Post-legislation							
β_{male}	0.058*** (7.84)	0.050*** (10.02)	0.048*** (9.68)	0.040*** (6.53)	0.050*** (14.32)	0.056*** (14.92)	0.074*** (8.54)
β_{non}	0.088*** (10.18)	0.095*** (16.35)	0.077*** (13.34)	0.085*** (12.08)	0.071*** (17.82)	0.072*** (16.46)	0.124*** (12.29)
R^2	0.774	0.783	0.775	0.788	0.769	0.780	0.750
N	69133	139529	141877	95173	326215	249675	55489
Mkt	Y	Y	Y	Y	Y	Y	Y
Firm	Y	Y	Y	Y	Y	Y	Y
Ing.	Y	Y	Y	Y	Y	Y	Y

$$p_{imt} = \beta_{male} male_{it} + \beta_{non} non_{it} + X_i' \alpha + \gamma_{mt} + \gamma_f + \epsilon_{imt}$$

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